Early-Stopping Self-Consistency

On a small open-weight model at a matched budget: fewer samples, no clear accuracy gain, more tokens

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Implementation and evaluation of the objective approved in Phase 1

The approved objective, and what was built

To evaluate early-stopping self-consistency (ESC; window 5, at most 20 samples) for arithmetic word problems on a fixed random subset of 50 questions from the GSM8K test set (1,319 questions), using the open-weight Qwen2.5-1.5B-Instruct model run locally, compared against fixed-size self-consistency (SC) at a matched sample budget, measured by exact-match accuracy and by samples, tokens, sequential rounds and wall-clock latency per question, each reported as mean and standard deviation over three seeded runs.

Built: greedy CoT and SC-k baselines, ESC, a random-stopping ablation, Adaptive-Consistency for context, a resumable sampler with full traces, an offline evaluator and a live demo.

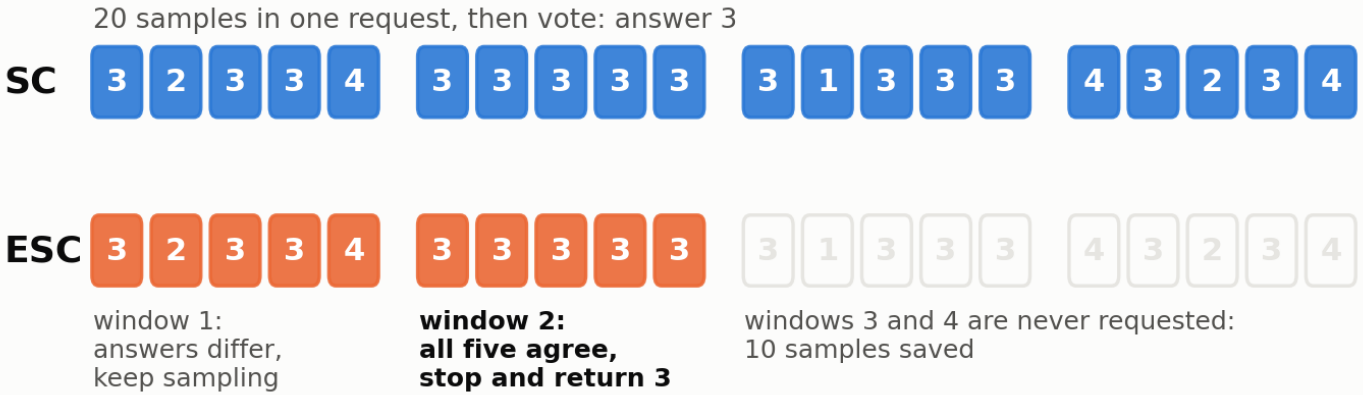
Scope: The objective was implemented as approved, with no change of scope.

Pinned configuration

Item	Value
Model	Qwen2.5-1.5B-Instruct, GGUF q4_k_m (sha256 6a1a2eb6...434e9407e)
Runtime	llama.cpp llama-server b11429, CPU only, 4 threads, 5 slots
Decoding	temperature 0.7, top-p 0.9, top-k off, repeat penalty 1.0, max 256 new tokens; greedy baseline at temperature 0
Prompt	system message + 3 worked examples of my own + question (prompts/, about 420 prompt tokens)
Tasks	GSM8K test, subset seed 465, 50 questions (data/subset_ids.json)
Runs	3, seeds 11, 22, 33; sample seed = run x 10^6 + question x 100 + index
Checker	exact match on the number after "Final answer", else the last number in the reply; no LLM judge
Date, cost	6 October 2026; 3000 sampled generations, 0.28M completion tokens, 3.1 h of CPU, zero rupees

Prompts are files in prompts/. Every run writes one JSON trace per question. Tables regenerate from the traces with no model call.

One pool of samples, so only the stopping rule differs



Sample once: 4 rounds of 5 per question and run.

SC-k: reads the first k samples.

ESC: reads whole rounds until one is unanimous.

Same: model, prompt, decoding, samples.

Matched budget

SC at ESC's mean sample count, rounded down (k = 15) and up (k = 16).

Ablation

Random stopping: ESC's sample counts shuffled across questions. Same budget, no signal.

Cost

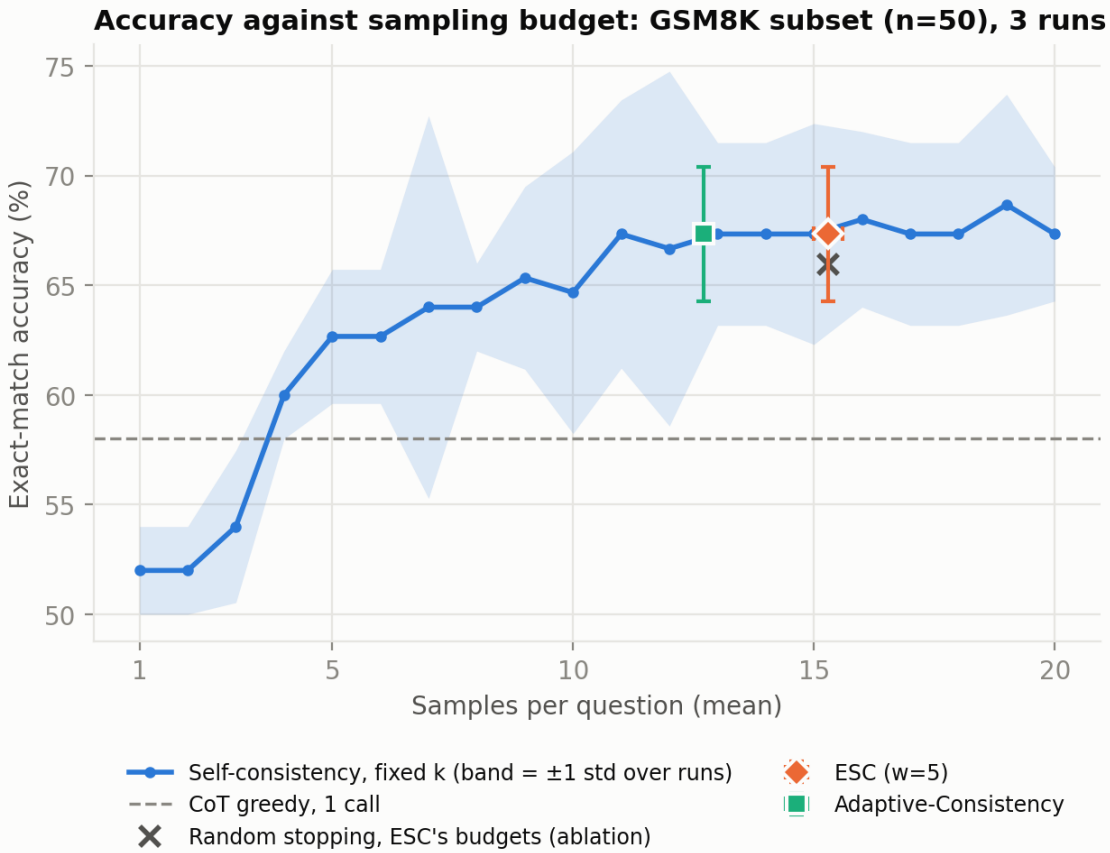
Samples, sequential rounds, prompt and completion tokens, measured wall-clock time per round.

Results: mean and standard deviation over 3 runs

Method	Accuracy %	Samples	Rounds	Prompt tok	Compl. tok	Total tok	Latency s
CoT greedy (1 call)	58.0	1	1	420	90	510	6.5
SC-5	62.7 ± 3.1	5.00	1.00	420	466	885	19.8 ± 2.4
SC-10	64.7 ± 6.4	10.00	1.00	420	918	1338	38.4 ± 3.7
SC-15 (matched, floor)	67.3 ± 5.0	15.00	1.00	420	1376	1795	56.6 ± 4.9
SC-16 (matched, ceil)	68.0 ± 4.0	16.00	1.00	420	1467	1886	60.2 ± 4.8
SC-20 (full budget)	67.3 ± 3.1	20.00	1.00	420	1836	2256	74.6 ± 4.7
ESC (w=5, L=20)	**67.3 ± 3.1**	**15.30**	**3.06**	**1295**	**1487**	**2782**	**60.7 ± 5.3**
Adaptive-Consistency	67.3 ± 3.1	12.71	12.71	5391	1257	6648	52.0 ± 5.9
Random stopping (ablation)	65.9 ± 3.4	15.30	3.06	1295			

50 GSM8K questions, costs per question. Prompt tokens: one prompt per round. Source: results/summary.csv, python src/evaluate.py.

Finding 1: no detectable accuracy gain at a matched budget



67.3%

ESC, 15.3 samples per question

68.0%

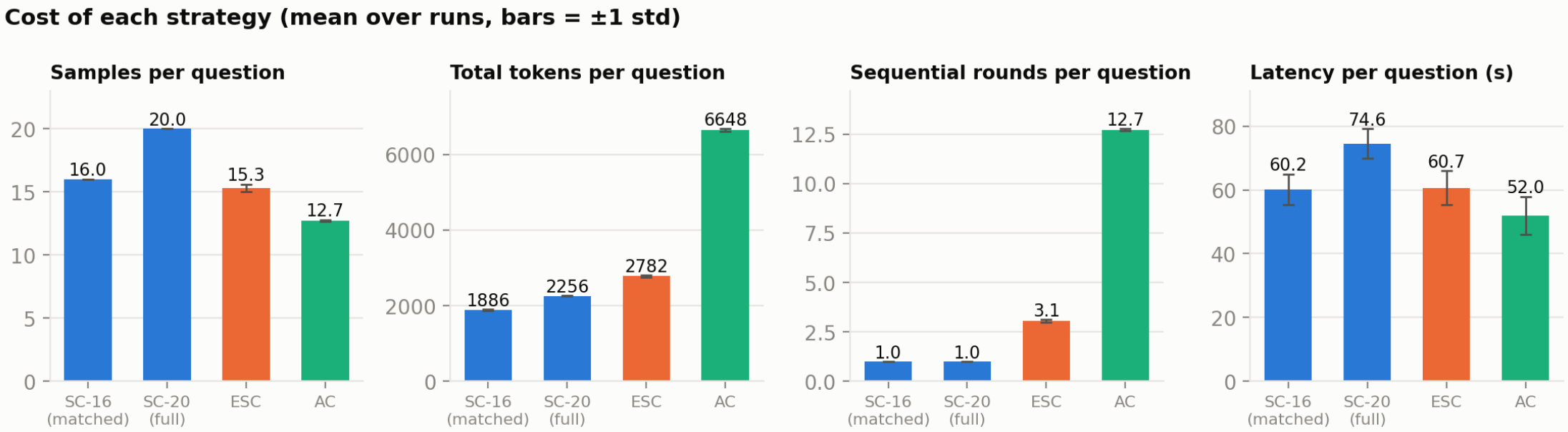
SC-16, the same budget

-0.7 ± 1.2

paired difference in points

Checker-sensitive: with replies that lack an answer line scored as no answer, ESC is ahead by +2.7 ± 1.2 points. Not settled at this sample size.

Finding 2: fewer samples, but more tokens and more rounds

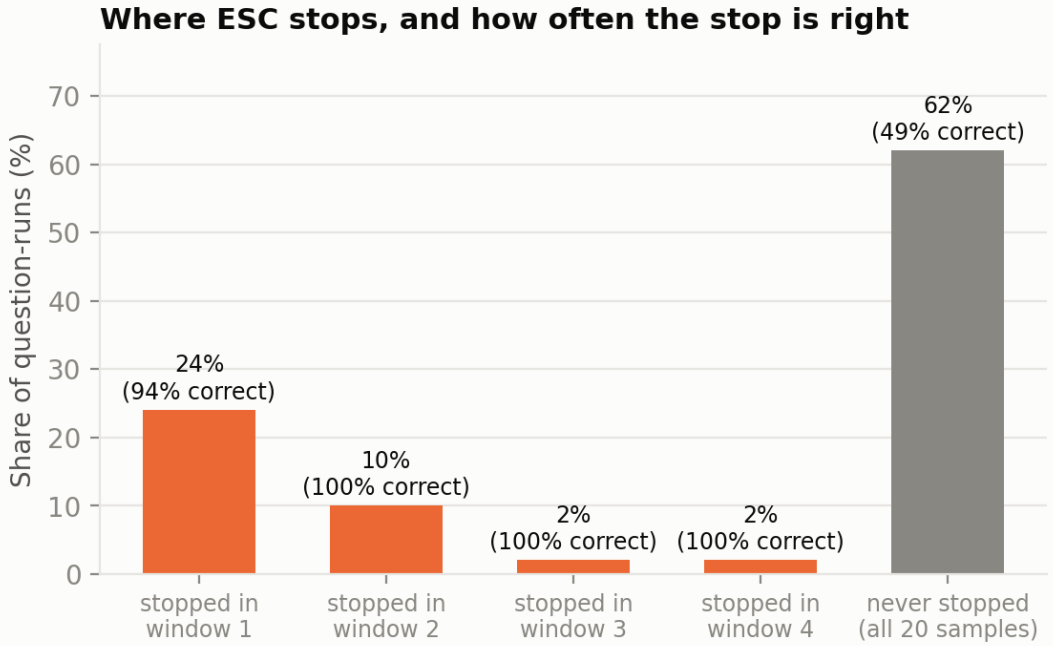


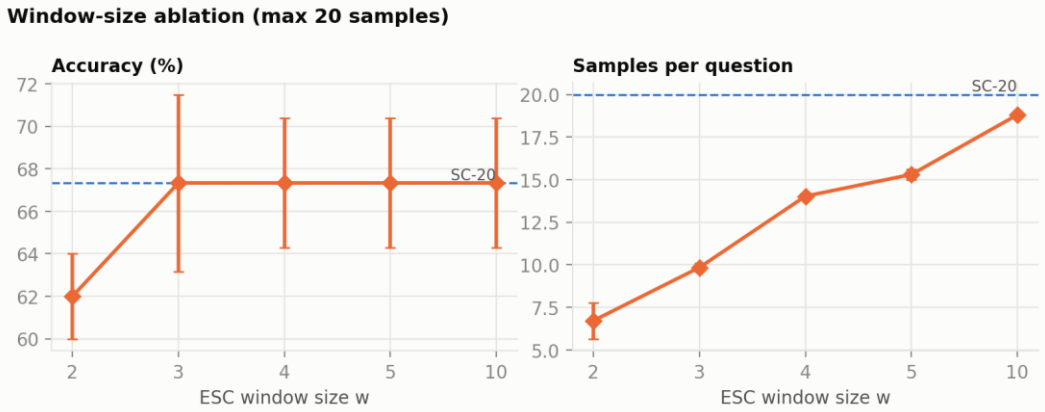
23% fewer samples than SC-20, with the same answers (agreement 100%).

23% more total tokens: every window re-sends the 420-token prompt.

3.1 sequential rounds against 1. Behind a parallel API the rounds are the latency.

Finding 3: the stopping signal is precise, and worth little





Random stopping with the same budgets: 65.9%. The signal is worth +1.4 ± 0.6 points.

Early stops: 57 of 150 question-runs; 2 of them wrong.

Window size: w = 2 uses 6.7 samples for 62.0%; w = 10 uses 18.8 for 67.3%.

A failure read from the trace: a confident wrong stop

Question 99

Mary receives 18 new potted plants. She already has 2 on each of her 40 window ledges. She gives away 1 plant from each ledge. How many plants does she keep?

Gold: 58

Run 11, first window

40 40 40 40 40

ESC stops after 5 samples and returns 40. Of all 20 samples, 18 say 40.

From the trace (traces/run_11/q0099.json)

*"Mary has 2 plants on each of 40 ledges, so she has 2 * 40 = 80 plants already. She will give away 1 plant per ledge ... 80 - 40 = 40."*

Cause: the 18 new plants in the first sentence are never used.

Unanimity measures consistency, not correctness.

No harm here: SC-20 returns 40 as well, so ESC lost nothing and saved 15 samples.

2 of 57 early stops were wrong; 12 of 50 questions are wrong in every run.

Limitations, next steps and the demo

Limitations

50 questions, one model, one prompt: only large effects are visible.

L = 20, not the paper's 40.

Greedy baseline 58% against 73.2% published for this model (4-bit weights, own prompt, small n).

Latency from a shared laptop; the round count is the portable figure.

With four more weeks

Full 1,319 questions and L = 40 on a GPU.

A second, stronger open-weight model.

Latency against a hosted API with parallel sampling.

ESC with the prompt cached across windows.

Live demo: python demo.py --task 6

ESC on one GSM8K question, window by window, with the trace written to traces/demo/.